Nifty 100 Financial Intelligence Platform

Analyst User Guide & API Reference Manual

1. System Overview & Architecture

The Nifty 100 Financial Intelligence Platform is an end-to-end institutional-grade analytics engine combining automated data pipelines, rule-based NLP extraction, machine learning clustering, a high-performance FastAPI REST backend, and an interactive Streamlit web dashboard.

2. Streamlit Dashboard Navigation

- **Home:** Macro-level market overview, index KPI summaries, and sector distribution charts.
- **Profile:** Deep dive into individual company balance sheets, ROCE/ROE trajectories, and P&L; histories.
- **Screener:** Custom multi-factor screening using real-time metric sliders and financial preset filters.
- **Peers:** Industry group peer benchmarking and interactive multi-axis radar comparison charts.
- **Trends:** Multi-metric overlay analysis for tracking YoY financial growth trajectories.
- **Clusters (New):** Machine learning KMeans financial archetypes ($k=5$), elbow curves, and correlation matrices.

3. Generating & Downloading PDF Tearsheets

Users can navigate to the **Reports** module inside the dashboard to instantly preview and download programmatically compiled 2-page company tearsheets and 11 sector intelligence summaries generated via ReportLab.

4. FastAPI REST API & Example cURL Commands

The backend server runs on FastAPI and exposes 16 documented endpoints. Interactive API documentation is available at http://localhost:8000/docs.

```
Health Check:
curl -X GET "http://localhost:8000/api/v1/health"

Fetch All Companies:
curl -X GET "http://localhost:8000/api/v1/companies?sector=IT"

Download Company Tearsheet PDF:
curl -X GET "http://localhost:8000/api/v1/companies/INFY/tearsheet" --output infy_tearsheet.pdf
```

5. Troubleshooting & Support

- *Port Conflicts:* Ensure port 8000 (FastAPI) and port 8501 (Streamlit) are free before launching.
- *Missing Data Errors:* Run the ETL ingestion script and clustering module (`python src/analytics/clustering.py`) if local artifacts are missing.